EEE-101 - Control Systems Engineering

Tutorial - 5 - Answers

1. Kept as an Exercise Problem.

2. Unstable; two poles with positive real parts.

3. $0 < K < \frac{21}{4}$.

4. 3 in left half plane; 2 in right half plane.

5. 2.

6. For unit impluse input: $e_{ss} = 0$; for unit step input: $e_{ss} = 0.9983$; and for unit ramp input: $e_{ss} = \infty$.

7. (a) The system is marginally stable.

(b) Steady state error = 0.

8. System type = 1; $K_p = -1.25$; $e_{ss} = -4$.

9. Error due to Reference Input = $\frac{1}{11}$; Error due to Disturbance Input = $-\frac{39}{11}$; Total Steady-State Error = $-\frac{38}{11}$.

10. $\frac{1}{a-1}$.